

BTEC Level 3 Applied Science • Unit 1 Biology • Answer sheet

Prokaryotic and Eukaryotic Cells — Answers

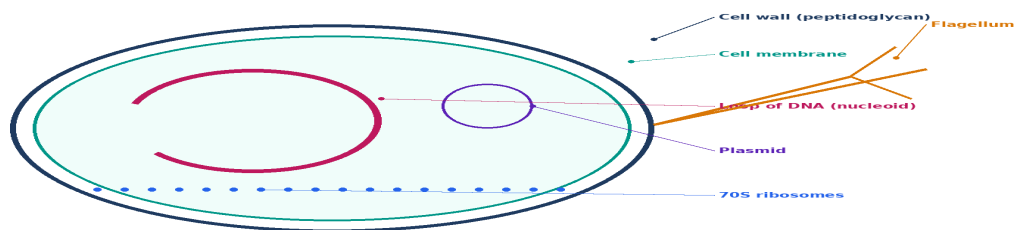
Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. Eukaryotes have a nucleus / membrane-bound organelles; prokaryotes do not.
2. Cell wall, membrane, loop of DNA, ribosomes; also plasmid / flagellum / capsule.
3. Neither — they are acellular / not cells.
4. Accept correct labels from the schematic.

Typical prokaryotic cell (not to scale)



5. Nucleus absent/present; circular vs linear DNA; no/yes membrane-bound organelles; 70S vs 80S; 1–5 μm vs 10–100 μm ; peptidoglycan vs cellulose/chitin/none.
6. Small extra DNA circle; may carry resistance genes or be used as a vector.
7. Antibiotics can target peptidoglycan walls or 70S ribosomes, which human cells lack or have in a different form.
8. Three paired differences. [3]
9. Carry resistance genes; transferable; used as vectors. [3]
10. No nucleus / DNA not enclosed by a nuclear envelope / no membrane-bound organelles. [2]
11. These features resemble free-living bacteria, supporting an engulfed-prokaryote origin.